Network Epistemology — Collaborator Brief

A self-contained description of the model, the network setting, and the Root Node Hypothesis. The aim is to give you (a) enough to reconstruct any simulation result from scratch, and (b) a clean target for analytic work on convergence and absorption probabilities.

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1. The Model

1.1 The two-armed bandit (the world)

There are two competing theories about a Bernoulli process:

- **Theory 0** ("bad"): true success rate $p_0 = 0.5$.
- **Theory 1** ("good"): true success rate $p_1 = 0.5 + \varepsilon$, with $\varepsilon \in (0, 0.5]$.

We call ε the **bandit gap** (the code calls it `uncertainty`). Typical values in our sims: $\varepsilon \in 0.001, 0.01, 0.05, 0.1$. Small ε is the interesting regime — it's hard to tell the arms apart.

"Truth" = Theory 1. An agent "believes the truth" when its credence on T_1 exceeds its credence on T_0.

1.2 Agents and their belief state

We have N agents arranged on a directed graph $G = (V, E)$ with $|V| = N$. Each agent maintains a Bayesian belief about **each** theory independently. We use Beta-Bernoulli conjugacy.

For each agent i and each theory $T \in 0, 1$, the belief is a Beta distribution:

$$\pi_i^T ; =; \mathrm{Beta}\big(\alpha_i^T ,; \beta_i^T\big).$$

The **credence** is the posterior mean:

$$c_i^T ; =; \frac{\alpha_i^T}{\alpha_i^T + \beta_i^T}.$$

(Optional toggle: instead of the mean, sample $c_i^T \sim \mathrm{Beta}(\alpha_i^T, \beta_i^T)$ each step. Default is the mean.)

1.3 Initial conditions

At $t = 0$, for every agent i and every theory T, the prior parameters are drawn independently:

$$\alpha_i^T,;\beta_i^T;\overset{\text{i.i.d.}}{\sim};\mathrm{Uniform}(0,4).$$

So each agent's initial credence on T_1 is roughly uniform over $(0,1)$, with a distribution that puts mass mainly near 0.5 but with non-trivial weight near the boundaries.

1.4 Choice rule

At each step, every agent picks **one** theory to test. We use **ε-greedy**:

$$T_i(t);=;\begin{cases}\arg\max_T c_i^T(t) & \text{with prob. } 1-\varepsilon_{\text{explore}}, \\[2pt]\text{uniform in } {0,1} & \text{with prob. } \varepsilon_{\text{explore}}.\end{cases}$$

Important: the default in our sims is $\varepsilon_{\text{explore}}=0$ (pure exploit). This is not a typo — we are deliberately studying the "lock-in" regime. (We use the symbol ε for the bandit gap; the exploration rate is a separate parameter, which we sometimes denote ε_e to disambiguate.)

1.5 The experiment

Agent i runs n independent Bernoulli trials of its chosen theory $T_i(t)$:

$$s_i(t);\sim;\mathrm{Binomial}!\left(n,;p_{T_i(t)}\right),\qquad f_i(t)=n-s_i(t).$$

The parameter n (`n_experiments` in code) is constant across agents and time. Typical values: $n\in{1,5,10,50}$.

1.6 Information sharing

This is where the network enters. Let $A\in{0,1}^{N\times N}$ be the adjacency matrix of G, with $A_{uv}=1$ iff $u\to v$. **The semantics is: $u\to v$ means "v observes u".** So v's predecessors ${u: A_{uv}=1}$ are the agents whose results v sees.

At step t, after experiments are run, agent v has access to its own outcome **plus** all predecessors' outcomes — but only the outcomes corresponding to the theory each agent actually tested. Crucially:

- If agent u tested T_0 at step t, then v sees (s_u,f_u) as evidence about T_0 only.
- v gets **no information about** T_1 from u at that step.

Define theory-indexed individual-outcome tensors $S_i^T(t),F_i^T(t)$ so that $(S_i^T,F_i^T)=(s_i,f_i)$ if $T_i(t)=T$ and $(0,0)$ otherwise. The aggregate evidence received by agent v for theory T at step t is

$$\widetilde{S}_v^T(t);=;S_v^T(t)+\sum_u A_{uv},S_u^T(t),\qquad \widetilde{F}_v^T(t);=;F_v^T(t)+\sum_u A_{uv},F_u^T(t).$$

(In matrix form: $\widetilde{S}^T=(I+A^\top)S^T$.)

1.7 Update rule

Beta-Bernoulli conjugate update, applied independently per theory:

$$\alpha_v^T(t+1); =; \alpha_v^T(t) + \widetilde{S}_v^T(t), \qquad \beta_v^T(t+1); =; \beta_v^T(t) + \widetilde{F}_v^T(t).$$

Then credences refresh: $c_v^T(t+1) = \alpha_v^T(t+1)/(\alpha_v^T(t+1) + \beta_v^T(t+1))$.

Structural fact (used a lot below): α_v^T, β_v^T are monotonically non-decreasing in t. The state (α, β) lives in $\mathbb{Z}_{\geq 0}^{N\times 2\times 2}$ shifted by the initial real priors; trajectories never revisit a state. The chain on this state space is **absorbing**: as $t \to \infty$, $\alpha + \beta \to \infty$ for any theory that gets tested infinitely often, so the credence on such a theory concentrates at the true rate (SLLN). An untested theory's credence stays at its initial value.

1.8 Convergence and "truth share"

We stop when consecutive credence vectors agree within tolerance:

$$|c(t) - c(t-1)|_\infty < \mathrm{tol}, \quad \mathrm{tol} = 5 \times 10^{-3}.$$

(Or at a fixed max step count, often 10^6.) Once stopped, the **truth share** is

$$\mathrm{TS}; =; \frac{1}{N}, \left|i : c_i^1 > c_i^0\right|.$$

This is the main outcome variable.

1.9 Parameters at a glance

Symbol	Meaning	Typical
N	network size	50–500
ε	bandit gap $p_1 - p_0$	$0.001–0.1$
ε_e	exploration rate	0 (default)
n	experiments per agent per step	$1–50$
α_0, β_0	initial Beta params	$\sim U(0, 4)$
$t_{\max}$	max steps	10^6
tol	stopping tolerance on credences	5×10^{-3}

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2. Networks

2.1 What we have

We generate or import three families:

1. **Synthetic generators** (in `networks/network_generation.py`):

    - **Directed Barabási–Albert** — preferential attachment on out-degree. Edges go from "cited" to "citing", which gives a heavy-tailed in-degree distribution and a handful of well-cited hubs.
    - **Directed Watts–Strogatz** — ring lattice with rewiring.
    - **Directed Erdős–Rényi** — i.i.d. edges.

2. **Empirical citation networks** (in `networks/citation_data/`):

    - **PUD** — Peptic Ulcer Disease research, 1900–1978, from OpenAlex. After pruning "twin" authors (always co-author, treated as one epistemic unit) and extracting the largest weakly-connected component, the network has $N = 312$ nodes and **72 roots**. It is a **DAG** (no cycles).
    - Tobacco, Ego (similar pipelines).

Edge convention is the same across families: $u \to v$ means v observes u (i.e., u is cited by v).

2.2 The case we want to focus on

We are focused on **directed networks with root nodes** — nodes with in-degree zero. In the PUD network roughly 23% of nodes are roots.

A root has a special epistemic status: it observes only its own experiments. So **a root is exactly an isolated bandit agent**, embedded in a larger graph. All the network structure does is propagate root beliefs to descendants. This is the structural intuition behind the hypothesis below.

In particular, when G is a **DAG** (the PUD case), every node is reached by some root, and the partition by "which roots can reach me" is well-defined.

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3. The Root Node Hypothesis (as a lower bound)

3.1 Statement

Let $R = r \in V : \deg^-(r) = 0$ be the set of roots. Let $R_{\text{true}}(t) \subseteq R$ be the subset of roots that, at time t, believe the truth ($c_r^1(t) > c_r^0(t)$). Let $\mathrm{Desc}(r)$ denote the set of nodes reachable from r along directed edges (including r itself). Define the **root-reachable truth share** at time t:

$$\mathrm{LB}(t); :=; \frac{1}{N}, \Big| \bigcup_{r \in R_{\text{true}}(t)} \mathrm{Desc}(r) \Big|.$$

Hypothesis (corrected framing — lower bound). As $t \to \infty$,

$$\mathrm{TS}(t); \geq; \mathrm{LB}(t) \quad \text{in the limit, i.e.} \quad \liminf_{t\to\infty} \big[\mathrm{TS}(t) - \mathrm{LB}(t)\big]; \geq; 0.$$

That is: **every descendant of a truthful root eventually converges to truth, but additional nodes may also converge to truth via mechanisms independent of root influence.** Root-reachability provides a baseline truth share, not the full picture.

The original ROOTNODE_HYPOTHESIS.md phrases this as the *entire* epistemic state being "determined by" root beliefs. That is too strong — descendant convergence is one source of truth, but not the only one. We are explicitly correcting that here.

3.2 Why it's plausible (heuristic), in two halves

Half 1 — root-driven truth (the lower bound itself).

1. **Roots are isolated bandits.** A root never receives evidence about a theory it does not personally test. Once it commits to one arm (in the $\varepsilon_e = 0$ regime), the other arm's belief is frozen at its prior mean. So a root's fate is determined by (i) initial priors, (ii) the random sequence of Bernoulli trials it runs on whichever arm it picks first.

2. **Downstream signal swamping.** For any descendant v of a truthful root r, evidence on T_1 accumulates at v linearly in t: per step, r contributes n Bernoulli trials at rate $p_1 = 0.5 + \varepsilon$, and the same goes for every truthful intermediate. By SLLN, $c_v^1(t) \to 0.5 + \varepsilon$ a.s.; meanwhile c_v^0 either stays at its prior or concentrates at 0.5. Either way $c_v^1 > c_v^0$ eventually.

This is what gives us "TS $\geq$ LB" — every descendant of a truthful root is counted in TS.

Half 2 — non-root truth (the excess).

A node v not downstream of any truthful root can still end up at $c_v^1 > c_v^0$. Plausible mechanisms:

- **Favorable-prior lock-in.** v's initial prior already has $c_v^1(0) > c_v^0(0)$ and the early evidence (own + neighbors') is consistent enough with T_1 that the agent stays committed to the correct arm, even though no truthful root is upstream to keep feeding it T_1 evidence.
- **Cycle-supported consensus.** In a strongly-connected component with no truthful root, agents can collectively reinforce T_1 beliefs by circulating their own T_1 experiments. (Not possible in strict DAGs.)
- **Isolated components.** Nodes in a weakly-connected component that contains no root in R_{true} but where individual bandit dynamics happen to land on T_1.

We don't expect these to dominate, but they're real, and they're why the relation is $\geq$ rather than $=$.

3.3 Empirical status (PUD, $N = 312$, 72 roots, $\varepsilon = 0.05$, n = default)

Running one realization for 10^6 steps:

Steps	$\mathrm{LB}(t)$ **(root-reachable share)**	**Actual** $\mathrm{TS}(t)$	$\mathrm{TS} - \mathrm{LB}$
10^3	0.7244	0.4455	-0.2788
10^4	0.7436	0.5994	-0.1442
10^5	0.7821	0.7821	;; 0.0000
10^6	0.7628	0.7724	$+0.0096$

Interpretation under the corrected framing:

- The **negative gap at early** t is the *lag phase*: the lower bound hasn't been achieved yet because signal hasn't propagated from roots to all their descendants. TS is *below* LB transiently — this is allowed because LB is the asymptotic lower bound, not a per-step one.
- The **positive gap at** 10^6 is the *excess from non-root mechanisms*. It is not an anomaly; it is exactly what the $\geq$ framing predicts.
- The exact equality at 10^5 is a snapshot coincidence on this single realization — across replicates we'd expect a small positive distribution.

3.4 Math we have / want

Let's pin down the natural mathematical objects.

(a) Single-root convergence probability. Fix an isolated agent with priors $(\alpha_0^0, \beta_0^0, \alpha_0^1, \beta_0^1) \in [0,4]^4$ uniform. In the $\varepsilon_e = 0$ regime, the agent commits at $t = 0$ to whichever theory has the higher prior mean, say T^*, and from then on:

- runs Bernoulli(p_{T^*}) trials,
- updates only $(\alpha^{T^}, |beta^{T^})$,
- leaves the other theory's parameters frozen.

So $c^{T^}(t) |to p_{T^}a.s.Theagentstayscommittedto T^iffc^{T^}(t) > c^{1-T^}_0forallt$. *Since the unchosen theory's credence is the* ***frozen prior mean*** *$m := |alpha^{1-T^}_0 / (\alpha^{1-*T^}_0 + |beta^{1-T^}_0), thequestionbecomes: doestherandomwalk c^{T^*}(t) evercrossbelow m$?

- If $T^* = 1$ (the agent initially picks the truth): $c^1(t) \to 0.5 + \varepsilon$. The walk may dip below m early, in which case it switches arms. **Open question:** compute $\mathbb{P}(\text{never switches} \mid T^* = 1)$ as a function of $(\alpha_0^1, \beta_0^1, m, \varepsilon, n)$.
- If $T^* = 0$: $c^0(t) \to 0.5$. Switching is more likely (the limit is exactly 0.5, which is at the typical scale of m).

A cleaner sub-question: under the uniform prior on $(0,4)^4$, what is the unconditional probability $q := \mathbb{P}(\text{root converges to truth})$? We have empirical estimates per network; we'd love a closed form or at least an integral expression.

(b) Descendant propagation lemma (the lower-bound half). Given a root r has converged to $c_r^1 > c_r^0$ (so α_r^1, β_r^1 are growing toward the empirical rates of T_1), prove that every descendant v of r eventually has $c_v^1 > c_v^0$ almost surely. Empirically this looks like 1 at any distance, but no formal proof.

A useful decomposition (revised from `OPEN_QUESTIONS.md` §4 under the lower-bound framing):

$$ \mathrm{TS}(\infty) ;=; \underbrace{\mathrm{LB}(\infty)}*{\text{root-driven}} ;+; \underbrace{X(\infty)}*{\text{non-root excess}}, \qquad X(\infty) \geq 0. $$

The lower-bound hypothesis is equivalent to "$\mathrm{LB}(\infty) \leq \mathrm{TS}(\infty)$ a.s." Lemma (b) above gives us the LB term. We then want to characterize $\mathbb{E}[X(\infty)]$ separately.

(c) The lag phase. $\mathrm{TS}(t) - \mathrm{LB}(t)$ starts strongly negative (signal hasn't propagated) and crosses zero somewhere between 10^4 and 10^5 steps in the PUD example. The lag should scale with some structural property of the graph — diameter? mean root-to-leaf path length? Spectral gap of the row-normalized adjacency matrix? We have no analytic bound.

(d) The non-root excess $X(\infty)$. Under the corrected framing this is not an anomaly but a quantity to model. Candidate mechanisms (any non-root truth-believer must come from one of these):

- A handful of non-root nodes have favorable priors and happen to test T_1 enough times in the first few steps to lock in correctly before any neighbor's signal arrives or before unfavorable evidence from non-truthful upstream nodes accumulates.
- Self-reinforcing cycles among strongly-connected subgraphs with no truthful root. (Not possible in strict DAGs — but the PUD network is a DAG, so this is ruled out for PUD specifically.)
- Nodes in components or subgraphs without any truthful root where individual bandit dynamics happen to land on T_1 despite upstream pressure toward T_0.

For PUD ($N = 312$, DAG) the $+0.96$ is ~ 3 extra agents. We'd like to characterize the distribution of $X(\infty)$ across independent realizations, and predict $\mathbb{E}[X(\infty)]$ from graph structure + parameters.

3.5 Open analytic questions we'd love your help with

In rough order of how excited we are:

1. **Single-root truth probability.** Compute $q = \mathbb{P}(\text{root converges to truth})$ for uniform $(0,4)^4$ priors, as a function of (ε, n), with $\varepsilon_e = 0$. Closed form or clean integral both welcome. This pins down $\mathbb{E}[|R_{\text{true}}(\infty)|]$.

2. **Descendant propagation lemma (the LB half).** Show formally that conditional on a root r being truthful, every descendant of r converges to truth a.s. (in the $\varepsilon_e = 0$, $n \geq 1$ regime).

What is the minimal condition under which this holds? This is what makes LB a valid lower bound at all.

3. **Non-root excess $X(\infty)$.** Characterize $\mathbb{E}[X(\infty)]$ — the contribution to TS from nodes *not* downstream of any truthful root. Even an upper bound (e.g., "no more than $f(\varepsilon, n, G)$ of the network can be truth-believers without a truthful upstream root") would be very useful.

4. **Lag-phase bound.** Bound $\mathbb{E}[\mathrm{LB}(t) - \mathrm{TS}(t)]_+$ from above as a function of t and a graph-structural parameter. We suspect graph diameter or some spectral object is the right knob.

5. **Phase transition in ε_e.** With non-zero exploration, the "commit at $t = 0$" picture breaks. Is there a critical ε_e^* below which roots have positive probability of locking onto T_0, and above which they almost surely converge to T_1?

6. **Generalization to cycles.** PUD is a DAG. For graphs with cycles (BA, WS, ER), "root" isn't always well-defined. Is there a natural surrogate? In-degree-1 nodes? Low-betweenness sources? Strongly-connected-component condensation? See §3.6 for our current best candidate (eigenvector centrality). In the cycle case, $X(\infty)$ in (3) becomes much more interesting because cycles can support consensus without a root.

3.6 Eigenvector centrality as a root surrogate

We suspect eigenvector centrality is the right generalization of "rootness" — both because it captures a similar intuition on DAGs and because it remains well-defined on graphs with no roots at all.

Definition. For a directed graph with adjacency matrix A (recall $A_{uv} = 1$ iff $u \to v$, i.e., v listens to u), the *influence* eigenvector centrality $x \in \mathbb{R}^N_{\geq 0}$ is the leading right eigenvector of A:

$$A, x; =; \lambda_{\max}, x, \qquad x_u; =; \frac{1}{\lambda_{\max}} \sum_{v,:,u \to v} x_v.$$

That is, u is influential to the extent that it points to influential listeners. (The dual, "receptivity" centrality, is the leading right eigenvector of $A^\top$ — high if you listen to influential sources.) Under Perron–Frobenius, $\lambda_{\max}$ is real and x has non-negative entries; the cleanest setting is when the graph (or the relevant component) is strongly connected, otherwise PageRank-style regularization $M = (1-d)A/\mathrm{out}(\cdot) + d\mathbf{1}\mathbf{1}^\top/N$ is the standard fix.

Why it should help here.

1. **Roots are the extreme case.** A root has no incoming edges, so its belief is set purely internally and then broadcast. In $A^\top$ terms, a root's row is zero. In A terms, its column is zero — but its **row of A** can be dense and feed into many high-centrality listeners. So roots typically sit at the high end of influence centrality x. The "root reachability" lower bound

implicitly weights every truthful root by $|\mathrm{Desc}(r)|/N$; eigenvector centrality is the spectral analogue of that weight, generalized to non-root nodes.

2. **Conjectured weighted lower bound.** A natural eigenvector-centrality version of the hypothesis: in any directed graph (with or without roots),

$$\mathrm{TS}(\infty);\gtrsim;\frac{\sum_u x_u, 1[c_u^1(\infty) > c_u^0(\infty)]}{\sum_u x_u}.$$

   When G is a DAG, the truthful-root nodes have outsized x_u and the bound reduces (approximately) to the root-reachability LB. When G has no roots, this still gives a meaningful weighted baseline: the agents whose locked-in belief carries the most spectral weight set the asymptotic truth share.

3. **Connection to mixing time.** The spectral gap $1 - |\lambda_2/\lambda_{\max}|$ of the row-normalized adjacency (the "listening matrix" $L = D^{-1}A^\top$ with D the in-degree matrix) governs how fast a perturbation in any single agent's belief propagates to the rest of the network. This is the same spectral object that would naturally appear in a lag-phase bound (§3.5 question 4).

Open analytic questions in this direction.

- (a) Does the weighted bound above actually hold? Under what conditions on G and parameters?
- (b) On strongly-connected graphs with no roots, what determines $\mathbb{P}(c_u^1(\infty) > c_u^0(\infty))$ for a fixed node u? We conjecture it depends on x_u (or on a centrality-weighted average of upstream truthfulness), but have not formalized.
- (c) Is the second eigenvalue λ_2 a quantitative predictor of the empirical mixing time we observe in simulation?

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4. References inside this repo (for context, not required)

- `model/vectorized_model.py` — the simulation engine (Beta agents, ε-greedy, network aggregation).
- `model/bandit.py` — the two-armed Bernoulli bandit.
- `model/convergence_analysis/root_node_influence/ROOTNODE_HYPOTHESIS.md` — the hypothesis statement with the PUD table reproduced above.
- `model/convergence_analysis/OPEN_QUESTIONS.md` — broader list of open conceptual questions, including the factorization conjecture in §3.4(b).
- `model/convergence_analysis/root_node_influence/root_influence_analysis.py` — the script that produced the PUD numbers above.

Happy to share any of these directly if useful.